

ARJUN AGRAWAL

Ghaziabad, India

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EDUCATION

Bachelor of Technology (B.Tech) – Computer Science Engineering KIET Group of Institutions, Ghaziabad (2024–2028)

Current Status:** Second-year undergraduate student - Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Artificial Intelligence, Java (OOPS)

DAV Public School, Pratap Vihar, Ghaziabad

- Class XII (2024) | Class X (2022)

EXPERIENCE

Social Internship – Accessible Tech Guides Project

- Identified real-world digital accessibility problems faced by senior citizens and non-technical users.
 - Created **bilingual (Hindi & English)** step-by-step guides for UPI apps, IRCTC ticket booking, WhatsApp features, government schemes, online medicine ordering, and telemedicine platforms.
 - Conducted needs assessment, gathered user feedback, and compiled a **4-page impact report** with daily logs and geo-tagged photos.
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PROJECTS

IntelliFlow – AI-Based Intelligent Traffic Signal System

- Designed a smart traffic management system using **YOLO-based vehicle detection** to estimate traffic density from live video feeds.
- Implemented **adaptive signal timing logic** to dynamically optimize traffic flow.
- Added **emergency vehicle preemption logic** to prioritize ambulances and fire services.
- Trained a custom deep learning model using **PyTorch with CUDA acceleration** (50 epochs) to understand model training workflows.

- Built a basic backend using **Flask** and conceptual dashboards for monitoring traffic and signal states.
- Focused on **system design, integration, and problem-solving**, rather than just model accuracy.

Accessible Tech Website

- Built a bilingual educational website to improve digital inclusion for non-technical users.
- Structured content with clear navigation and simplified explanations.
- Emphasized usability, readability, and real-world applicability.

CivicEye – AI-Assisted Civic Issue Reporting (Concept + Prototype)

- Designed a **crowdsourced civic reporting system** allowing citizens to report issues with photos and location data.
- Incorporated **AI-driven verification** to filter harmful media, detect spam/duplicate reports, and assign confidence scores to reduce fake complaints.
- Proposed **heatmap-based visualization** of reported issues to highlight high-density and high-priority areas for faster decision-making.
- Introduced a **citizen-verified resolution flow**, where issues are marked resolved only after approval by a majority (>50%) of reporting users.
- Added **gamified engagement mechanisms** to incentivize responsible reporting and sustained citizen participation.

CareerCraft – AI-Powered Student Career Planner (Entrepreneurial Project)

- Developed the concept for an ERP-integrated platform that generates **personalized student career roadmaps**.
- Designed features for skill-gap analysis, milestone tracking, gamification (XP & badges), and auto CV generation.
- Created and presented a pitch deck; project won an **Internal Entrepreneurship Event**.

OTHER PROBLEM-SOLVING IDEAS & SYSTEM THINKING

- Genre-based e-commerce platform for **anime, pop culture, movies, K-dramas, and gaming merchandise**, focusing on niche communities, creator collaborations, and sustainable production models.
- AI-powered **smart glasses concept** for real-time assistance such as navigation hints, object recognition, or contextual overlays for daily tasks.
- **AI Vlogging Mate** for moto-vloggers: an intelligent system that automatically captures highlights, adjusts camera angles, and selects key moments based on speed, sound, and scene detection.

- Content recommendation and discovery systems for fandom-driven communities (anime, movies, music) to surface niche and underrated content.
 - Creator-first platforms that simplify editing, publishing, and monetization for short-form and long-form digital content using AI assistance.
 - Exploring how AI tools can augment creativity rather than replace it, especially in media, storytelling, and community-driven platforms.
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TECHNICAL SKILLS

Programming & CS Fundamentals

- Python, C, C++, Java
- Strong foundation in problem-solving, logic building, and OOPS concepts

Web & Backend Fundamentals

- HTML, CSS, JavaScript (basic)
- Backend concepts with Flask
- Databases

AI / ML Exposure

- PyTorch (model training basics, CUDA usage)
- NumPy, Pandas, Scikit-learn (introductory)
- Understanding of model training workflows and evaluation

Developer Tools & Platforms

- AI & Dev Tools: Windsurf, Cursor, Antigravity, Warp
 - AI APIs & Models: Groq, OpenRouter, Hugging Face models (API-level usage)
 - Design & Planning: Eraser.io
 - Deployment & Backend Services: Railway, Vercel, Supabase
 - Auth & Infra: Clerk
 - Version Control & CI/CD: Git, basic CI/CD concepts with Git + Vercel
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CERTIFICATIONS

- Introduction to Artificial Intelligence – LinkedIn Learning
 - AI Developer Series – Coursera
 - Java Object-Oriented Programming – LinkedIn Learning
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ACHIEVEMENTS

- 1st Year cgpa - 9.27
 - Winner – Internal Entrepreneurship Event (CareerCraft)
 - Internal Smart India Hackathon: Top 45 out of 610+ teams (Civic-Eye)
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LANGUAGES

- Hindi (Native)
 - English (Fluent)
 - Japanese (Beginner – basic reading, writing
Intermediate- speaking)
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PROFILE SUMMARY

Second-year Computer Science Engineering student with strong interest in **problem-solving, system design, and applied AI**. Experienced in building socially impactful and exploratory projects using guided resources and modern developer tools. Actively strengthening core computer science fundamentals while experimenting with AI, backend systems, and real-world applications.